

◆ SLEEPSCRIBE

Website Guide

AI Sleep Journal & Dream Analyst

A complete walkthrough of every feature,
the recommended demo order, and the system design.

Prepared for the Conesta Forge internship · React + Express + MongoDB + Groq

1. What is SleepScribe?

SleepScribe is an **AI-powered sleep journal and dream analyst**. People record their dreams — by text or by voice — and an AI model decodes each dream into a clear interpretation: a short summary, the emotional mood, the key symbols and themes, and a gentle sleep tip. Over time it surfaces patterns: recurring symbols, mood trends, journaling streaks, how sleep length relates to dream tone, and a weekly “sleep-coach” report.

It is a full-stack web app: a polished, animated marketing site that anyone can visit, plus a secure, login-protected app where each user keeps their private dream journal.

The product in one line

“Record your nights, decode your dreams, and wake up to smarter sleep.”

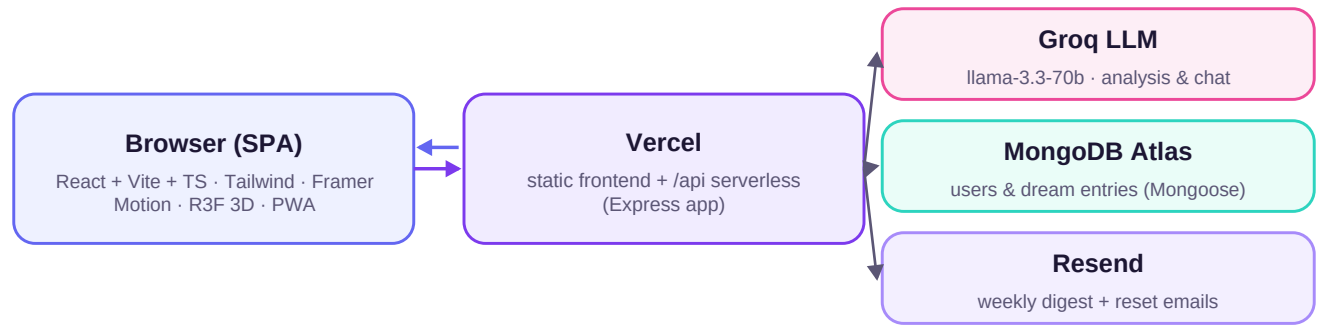
Who it is for

- **Anyone curious about their dreams** — instant, judgement-free interpretation.
- **People improving their sleep** — streaks, trends and a weekly report build the habit.
- **Visitors** — can try the dream AI and chat with the assistant without signing up.

2. How it is built (tech stack)

Layer	Technology	Purpose
Frontend	React 18, Vite, TypeScript, Tailwind CSS	Fast SPA + utility styling
Motion / 3D	Framer Motion, React Three Fiber, Lenis	Scroll animation & cinematic hero
Backend	Node.js + Express (serverless on Vercel)	REST API under /api
Database	MongoDB Atlas + Mongoose	Users & dream entries
AI	Groq SDK — llama-3.3-70b	Dream analysis, Luna, digests
Auth	JWT + bcrypt + Google Identity	Email/password & Google sign-in
Email	Resend API	Password reset & weekly digest
Extras	PWA (installable), Web Speech API	Offline shell & voice journaling

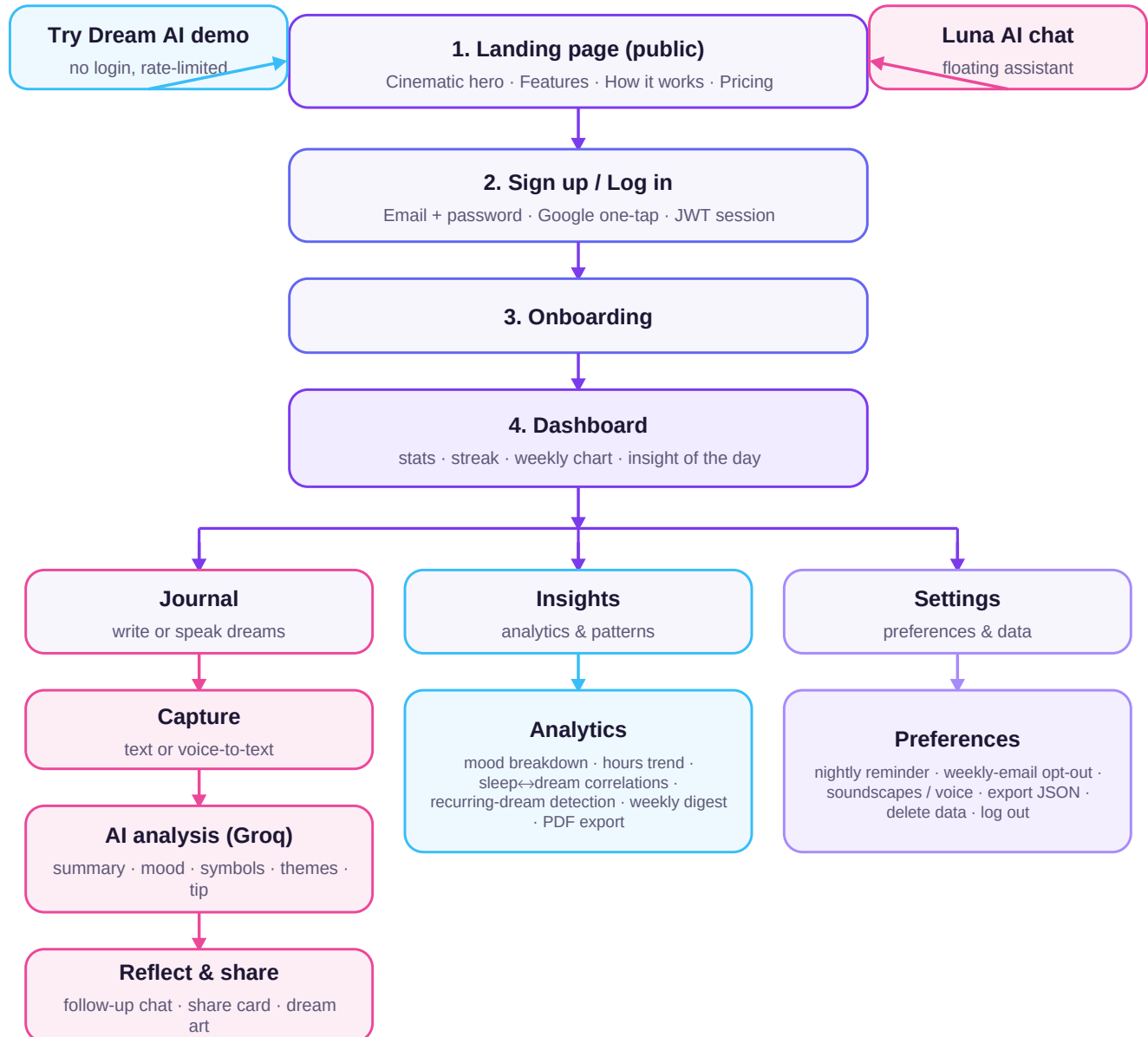
System architecture



JWT auth on every protected request · Groq key never leaves the server

3. How the website flows (workflow diagram)

The diagram below is the end-to-end user journey — from the public landing page, through sign-in, into the private app, and across the three core areas (Journal, Insights, Settings).



4. How to explain / demo the website (in order)

Follow this order when presenting. It moves from what anyone can see, to signing in, to the full private experience — the same path a new user takes.

- 1 **Open the landing page**
Start at the home page. Scroll slowly — the cinematic hero animation plays frame-by-frame as you scroll. Point out the value proposition, the feature grid, “How it works” (3 steps), testimonials and pricing.

- 2 **Try the Dream AI (no login)**
Scroll to the “Try the dream AI” section. Type or pick an example dream and click Analyze — a real AI interpretation appears live (summary, mood, symbols, themes, tip). This proves the AI works before anyone signs up.

- 3 **Chat with Luna**
Open the floating moon button (bottom-right). Ask a question like “What does dreaming of water mean?” Luna answers instantly. (Once logged in, Luna also reads your own journal.)

- 4 **Sign up / Log in**
Create an account with email + password, or use Google one-tap. Mention secure JWT sessions and the password-reset email flow.

- 5 **Land on the Dashboard**
After login you arrive at the dashboard: headline stats (avg sleep, streak, sleep score, entries), a weekly quality chart, the “insight of the day”, and recent dreams. Note the “Return to home” button.

- 6 **Write a dream in the Journal**
Go to Journal. Type a dream — or tap Speak to dictate it by voice. Click Save & analyze; the AI decodes it. Open the entry to see the interpretation, then use the follow-up chat, the share card, or “Visualize as art”.

- 7 **Explore Insights**
Open Insights: recurring-dream detection (with a Luna reflection), the weekly AI digest, mood breakdown, hours-slept trend, sleep↔dream correlations, and recurring symbols. Use “Export PDF” to download the whole journal.

- 8 **Adjust Settings**
Show the profile, the nightly reminder (real browser notification), the weekly-email opt-out, and data controls (export JSON / delete all). Finish with Log out.

5. Every feature, explained

5.1 Public landing experience

Cinematic scroll hero

What	A 240-frame video sequence scrubs as you scroll, with timed text overlays (“Entering REM”, etc.).
In app	Home page top section. Shorter and address-bar-safe on mobile.

Marketing sections

What	Feature grid, 3-step “How it works”, testimonials, and pricing tiers.
In app	Scroll the landing page; the navbar links jump to each section.

Live Dream AI demo

What	Visitors analyze a dream with the real AI — no account needed (rate-limited, with an offline fallback).
In app	“Try the dream AI” section → type a dream → Analyze.

5.2 Accounts & security

Sign up & log in

What	Email + password (hashed with bcrypt) or Google one-tap. Sessions use signed JWT tokens (7 days).
In app	/login and /onboarding pages.

Password reset

What	Request a reset link by email (via Resend); the link is valid for one hour.
In app	“Forgot password” → email link → reset page.

Protected app

What	All journal data is private and only loads for the logged-in owner.
In app	Everything under /app requires a valid session.

5.3 Dream journaling

Text & voice capture

What	Write a dream, or dictate it with live speech-to-text.
In app	Journal → New entry → type, or tap “Speak”.

AI dream analysis

What	On save, the AI returns a summary, mood (label + score + colour), 2-4 symbols, 2-4 themes, and a tip. Falls back to an offline analyzer if the AI is unavailable.
In app	Automatic on Save & analyze; shown in the entry detail.

Follow-up chat about a dream

What	Ask the AI questions about one specific dream in a threaded conversation.
In app	Open an entry → “Ask about this dream”.

Search & filter

What	Find dreams by keyword, mood, or time range (today / week / month).
In app	Journal → search bar + mood/time dropdowns.

5.4 Luna — the AI assistant

Public guide mode

What	A floating chatbot that answers questions about dreams, sleep and how to use the app.
In app	Moon button, bottom-right of every page.

Journal-aware mode

What	When you are signed in, Luna reads your own dream history and answers personally — recurring symbols, mood shifts, patterns across entries.
In app	Same button; shows “reading your journal” when logged in.

5.5 Insights & analytics

Dashboard overview

What	Avg sleep, journaling streak, sleep score and entry count — with real week/month deltas — plus a weekly chart, the insight of the day and recent dreams.
In app	/app home.

Recurring-dream detection

What	Finds the symbol or theme that recurs most and offers a Luna reflection grounded in those entries.
In app	Insights → top card.

Sleep ↔ dream correlations

What	Average hours slept by dream mood, and which symbols cluster on your shorter nights.
In app	Insights → correlations panel.

Weekly AI digest

What	A warm “sleep-coach” summary of the last 7 days of dreams.
In app	Insights → weekly digest. Also emailed (opt-out).

Charts & consistency

What	Mood breakdown, hours-slept trend, recurring symbols, and a real journaling-consistency score.
In app	Insights page.

5.6 Sharing, art & export

Share card

What	Generates a polished aurora-themed image of a dream’s analysis to share or download.
In app	Entry → “Share this dream”.

Dream art

What	Turns a dream into a unique generative aurora artwork, seeded from its text + mood.
In app	Entry → “Visualize as art”.

Export

What	Download the whole journal as a formatted PDF, or export raw entries as JSON.
In app	Insights → Export PDF; Settings → Export entries.

5.7 Engagement & settings

Streaks & nightly reminder

What	Tracks consecutive journaling days and can send a browser notification at a chosen time.
In app	Dashboard streak; Settings → Nightly reminder.

Weekly email digest + opt-out

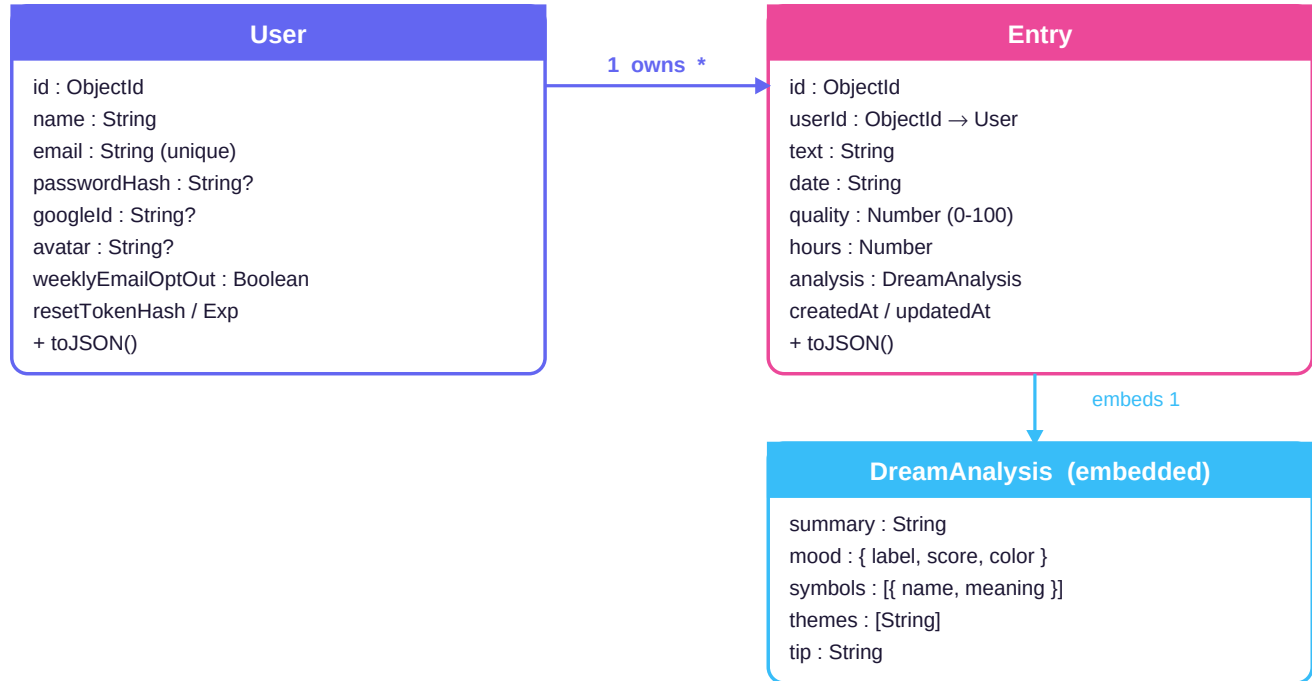
What	A scheduled job emails each active user their weekly report every Sunday; users can switch it off.
In app	Settings → Weekly AI report email.

Data controls

What	Export or permanently delete all journal data.
In app	Settings → Your data.

6. Data model (class diagram)

Two core collections back the app. A **User** owns many **Entry** documents; each entry embeds a **DreamAnalysis** object produced by the AI.



Relationships

- **User 1 — * Entry**: every dream entry stores the owner's `userId`; a user has many entries.
- **Entry 1 — 1 DreamAnalysis**: the AI result is embedded directly in the entry (no separate collection).
- Passwords are never returned to the client — `toJSON()` strips the hash and reset tokens.

7. Page & API map

Pages

Route	Page	Access
/	Landing (marketing site)	Public
/onboarding	Sign-up flow	Public
/login	Log in (email + Google)	Public
/forgot, /reset	Password reset	Public
/app	Dashboard	Protected
/app/journal	Dream journal	Protected

Route	Page	Access
/app/insights	Insights & analytics	Protected
/app/settings	Settings	Protected

Key API endpoints

Endpoint	Purpose
POST /api/auth/register, /login, /google	Create account / sign in
POST /api/ai/demo	Public dream analysis (rate-limited)
POST /api/ai/assistant	Luna chat (public)
POST /api/ai/journal	Luna chat grounded in your dreams (auth)
POST /api/ai/analyze, /chat	Analyze a dream / follow-up chat (auth)
GET /api/ai/digest	Weekly sleep-coach report (auth)
GET/POST/DELETE /api/entries	Manage dream entries (auth)
GET /api/cron/weekly-digest	Scheduled weekly emails (secured)

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